

Analytical
function

Window Function

(32)

Employee table

(check) Lin - sal)

EmpId FirstName LastName Age Salary Location

temp table

[written query and
created this table]

Location Total Avg-Salary
Employee in
Location

Group By

There is a chance when people ask to display non-aggregate columns also, when we are using GROUP BY. So at this point GROUP does not function properly.
So we use

✓ PARTITION BY

- * → It is similar to Group By
- * → But it allows Non-aggregated col also to display

SYNTAX

(33)

Select FN, LN, Location,

COUNT(Location) OVER (PARTITION BY Location)

AS Total

AVG (Salary) OVER (PARTITION BY Location)

AS Avg Salary

From Employee;

grouping happens

Row-Numbers () vs Rank () - vs

Dense Rank ()

Row Number Looking forward to create a new column, in that col employee with higher salary get their respective priority

ROW-NUMBER() → it assign unique values (B*)

map the values with numbers on the basis of sorted values of data that we have (ORDER BY)

* → whenever we use ROW-NUMBER(), always use ORDER BY() clause

Solution

SELECT FirstName, LastName, Salary,

{ ROW-NUMBER() OVER(ORDER BY Salary DESC)
AS Priority } → The new column
FROM Employee

It is not correct, when two ppl have same salary. It gives diff priority for same salary ppl.

RANK()

SELECT FN, LN, salary

{ RANK() OVER (ORDER BY salary
AS priority) } FROM Employee

The issue is resolved. But Priority 2 skipped, It directly jumped to 3

For ex

Monisha Periyasamy	200000	1
Monisha Sivakumar	200000	2
Priyadharshini Mariyappan	150,000	3
Abarna Sri	100000	4

DENSE_RANK()

SELECT FN, LN, salary

DENSE_RANK()

...

INTERVIEW QUESTION

ROW_NUMBER() VS RANK() VS
DENSE_RANK()

→ If there is no duplicate entries in your data, then ROW_NUMBER() VS RANK() VS DENSE_RANK() return the same result

INTERVIEW QUESTION (LeetCode)

178. RankScore